

Assignment 6 Submission

Advanced HTML5 & CSS3 Layouts

Flexbox, Grid, and CSS Animations

Theoretical Analysis & Layout Space Calculations

Before implementing the layouts, it is crucial to analyze the theories on different sides of CSS layout modules (Flexbox vs. CSS Grid) and how they calculate space.

- **Flexbox Theory (1-Dimensional Side):** Flexbox is fundamentally designed for 1D layouts (either a row or a column).

Calculation: Available Space = (Container Width) - (Sum of base widths of all flex items). If `flex-grow: 1` is applied, the browser calculates the remaining space and distributes it equally among the growing elements.

- **CSS Grid Theory (2-Dimensional Side):** Grid dictates both rows and columns simultaneously.

Calculation: When using Fractional Units (`fr`), the Grid calculates available space as: Available Space = (Container Width) - (Fixed column widths + Grid Gaps). If we declare `grid-template-columns: repeat(4, 1fr);`, the engine calculates 4 equal fractional spaces from the remaining width, meaning each column occupies exactly 25% of the flexible space.

1. Personal Portfolio Webpage Layout (Flexbox)

INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <div class="name">John Doe</div>
    <nav>
      <a href="#">Home</a>
      <a href="#">Portfolio</a>
    </nav>
  </header>
  <main>
    <section class="about">
      <h2>About Me</h2>
      <p>I am a web developer specializing in responsive design.</p>
    </section>
    <section class="image-section">
      
    </section>
  </main>
  <footer>
    <p>&copy; 2026 John Doe</p>
  </footer>
</body>
</html>
```

STYLE.CSS

```

body {
  margin: 0;
  font-family: sans-serif;
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}
header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 20px;
  background: #333;
  color: white;
}
main {
  display: flex;
  flex: 1;
  justify-content: center;
  align-items: center;
  padding: 20px;
}
.about, .image-section {
  flex: 1;
  padding: 20px;
}
footer {
  text-align: center;
  padding: 10px;
  background: #222;
  color: white;
}
@media (max-width: 700px) {
  main {
    flex-direction: column;
  }
}

```

2. 4-Column Product Gallery (CSS Grid)

[INDEX.HTML](#)

```
<main class="gallery">
  <!-- Repeat this card 8 times -->
  <div class="product-card">
    <h3>Product Name</h3>
    <p>$49.99</p>
  </div>
</main>
```

STYLE.CSS

```
.gallery {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 20px;
  padding: 20px;
}
.product-card {
  background: #f4f4f4;
  padding: 20px;
  border: 1px solid #ddd;
  text-align: center;
  transition: transform 0.3s ease;
  height: 100%; /* Ensures equal height */
  box-sizing: border-box;
}
.product-card:hover {
  transform: scale(1.05);
}
@media (max-width: 900px) {
  .gallery { grid-template-columns: repeat(2, 1fr); }
}
@media (max-width: 600px) {
  .gallery { grid-template-columns: 1fr; }
}
```

3. 3-Section Homepage (Flexbox)

INDEX.HTML

```
<div class="wrapper">
  <header class="site-header">
    <nav><a>Home</a><a>About</a><a>Contact</a></nav>
  </header>
  <main class="features">
    <div class="box">Services</div>
    <div class="box">Projects</div>
    <div class="box">Contact</div>
  </main>
  <footer class="site-footer">Footer Content</footer>
</div>
```

STYLE.CSS

```
.wrapper {
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}
.site-header nav {
  display: flex;
  justify-content: space-around;
  padding: 20px;
  background: #eee;
}
.features {
  display: flex;
  justify-content: center;
  gap: 30px;
  flex: 1;
  padding: 40px;
}
.box {
  background: #007bff;
  color: white;
  padding: 40px;
  transition: transform 0.3s;
}
.box:hover {
  transform: rotate(2deg);
}
.site-footer {
  width: 100%;
  background: #333;
  color: white;
  text-align: center;
  padding: 15px;
}
```

4. Dashboard Layout (CSS Grid)

[INDEX.HTML](#)

```
<div class="dashboard">
  <header class="dash-header">Header</header>
  <aside class="sidebar">Nav</aside>
  <main class="dash-main">
    <div class="dash-box">Box 1</div>
    <div class="dash-box span-2">Box 2 (Spans 2)</div>
    <div class="dash-box">Box 3</div>
    <div class="dash-box">Box 4</div>
  </main>
</div>
```

STYLE.CSS

```

.dashboard {
  display: grid;
  grid-template-columns: 250px 1fr;
  grid-template-rows: 60px 1fr;
  height: 100vh;
}
.dash-header {
  grid-column: 1 / -1;
  background: #2c3e50;
  color: white;
}
.sidebar {
  background: #34495e;
  color: white;
}
.dash-main {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 20px;
  padding: 20px;
}
.span-2 {
  grid-column: span 2;
}
.dash-box {
  background: #ecf0f1;
  padding: 20px;
  transition: transform 0.3s, box-shadow 0.3s;
}
.dash-box:hover {
  transform: scale(1.02);
  box-shadow: 0 10px 20px rgba(0,0,0,0.15);
}

```

5. Pricing Section (Flexbox)

[INDEX.HTML](#)

```
<section class="pricing-section">
  <div class="pricing-card">Basic</div>
  <div class="pricing-card highlight">Pro</div>
  <div class="pricing-card">Enterprise</div>
</section>
```

STYLE.CSS

```
.pricing-section {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 20px;
  padding: 50px 20px;
  animation: fadeIn 1s ease-in;
}
.pricing-card {
  background: white;
  border: 1px solid #ddd;
  padding: 40px;
  transition: transform 0.3s ease;
}
.pricing-card.highlight {
  background: #e3f2fd;
  transform: scale(1.1);
  padding: 50px 40px;
}
.pricing-card:hover {
  transform: translateY(-10px);
}
.pricing-card.highlight:hover {
  transform: scale(1.1) translateY(-10px);
}
@keyframes fadeIn {
  0% { opacity: 0; }
  100% { opacity: 1; }
}
@media (max-width: 750px) {
  .pricing-section {
    flex-direction: column;
  }
  .pricing-card.highlight {
    transform: scale(1);
  }
}
```

6. Responsive Blog Layout (CSS Grid)

INDEX.HTML

```
<div class="blog-layout">
  <header class="blog-header">Blog Title</header>
  <aside class="blog-sidebar">Categories</aside>
  <main class="blog-main">
    <article>Post 1</article>
    <article>Post 2</article>
  </main>
  <footer class="blog-footer">Contact Info</footer>
</div>
```

STYLE.CSS

```

.blog-layout {
  display: grid;
  grid-template-areas:
    "header header"
    "sidebar main"
    "footer footer";
  grid-template-columns: 250px 1fr;
  gap: 20px;
  padding: 20px;
}
.blog-header { grid-area: header; }
.blog-sidebar { grid-area: sidebar; }
.blog-main {
  grid-area: main;
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 20px;
}
.blog-footer { grid-area: footer; }
@media (max-width: 800px) {
  .blog-layout {
    grid-template-areas:
      "header"
      "main"
      "sidebar"
      "footer";
    grid-template-columns: 1fr;
  }
}

```

7. CSS Loading Animation

INDEX.HTML

```

<div class="loader-container">
  <div class="circle"></div>
  <div class="circle delay-1"></div>
  <div class="circle delay-2"></div>
</div>

```

STYLE.CSS

```
.loader-container {
  display: flex;
  justify-content: center;
  align-items: center;
  height: 200px;
  gap: 10px;
}
.circle {
  width: 20px;
  height: 20px;
  background-color: #3498db;
  border-radius: 50%;
  animation: bounce 0.6s infinite alternate;
}
.delay-1 { animation-delay: 0.2s; }
.delay-2 { animation-delay: 0.4s; }

@keyframes bounce {
  0% { transform: translateY(0); }
  100% { transform: translateY(-20px); }
}
```

8. Photo Gallery (Grid with Hover Transformations)

INDEX.HTML

```
<div class="photo-grid">
  <figure>
    
    <figcaption>Caption text</figcaption>
  </figure>
  <!-- Repeat figures -->
</div>
```

STYLE.CSS

```
.photo-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 15px;
}
figure {
  margin: 0;
  overflow: hidden;
  position: relative;
}
figure img {
  width: 100%;
  height: 100%;
  object-fit: cover;
  display: block;
  transition: transform 0.5s ease;
}
figure:hover img {
  transform: scale(1.1);
}
figcaption {
  text-align: center;
  padding: 10px;
  background: #f8f9fa;
}
```

9. Card Flip Animation

[INDEX.HTML](#)

```
<div class="flip-container">
  <div class="flip-card">
    <div class="flip-front">
      <h3>Title</h3>
      
    </div>
    <div class="flip-back">
      <p>Text description here.</p>
    </div>
  </div>
</div>
```

STYLE.CSS

```
.flip-container {
  display: flex;
  justify-content: center;
  align-items: center;
  perspective: 1000px;
  height: 300px;
}
.flip-card {
  width: 250px;
  height: 300px;
  position: relative;
  transform-style: preserve-3d;
  transition: transform 0.6s ease;
}
.flip-container:hover .flip-card {
  transform: rotateY(180deg);
}
.flip-front, .flip-back {
  position: absolute;
  width: 100%;
  height: 100%;
  backface-visibility: hidden;
  border: 1px solid #ccc;
  box-sizing: border-box;
}
.flip-back {
  transform: rotateY(180deg);
  background: #eee;
}
```

10. Responsive Webpage (Flexbox & Grid Combo)

INDEX.HTML

```
<div class="combo-layout">
  <header class="combo-header">
    <div class="logo">Logo</div>
    <nav>Links</nav>
  </header>
  <main class="combo-main">
    <article class="combo-article">
      <div class="inner-box animate-up">Box 1</div>
      <div class="inner-box">Box 2</div>
      <div class="inner-box">Box 3</div>
    </article>
    <aside class="combo-sidebar">Sidebar</aside>
  </main>
  <footer class="combo-footer">Centered Footer</footer>
</div>
```

STYLE.CSS

```
.combo-header {
  display: flex;
  justify-content: space-between;
  padding: 20px;
  background: #333;
  color: white;
}
.combo-main {
  display: grid;
  grid-template-columns: 3fr 1fr;
  gap: 20px;
  padding: 20px;
}
.combo-article {
  display: flex;
  gap: 15px;
}
.inner-box {
  flex: 1;
  background: #e2e8f0;
  padding: 20px;
}
.combo-footer {
  width: 100%;
  text-align: center;
  padding: 20px;
  background: #333;
  color: white;
  box-sizing: border-box;
}
.animate-up {
  animation: moveUp 1s ease-out;
}
@keyframes moveUp {
  0% { transform: translateY(30px); opacity: 0; }
  100% { transform: translateY(0); opacity: 1; }
}
@media (max-width: 768px) {
  .combo-main {
    grid-template-columns: 1fr;
  }
  .combo-article {
    flex-direction: column;
  }
}
```